



**QUESTIONING
REALITY**

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WHITE PAPER

Formulating Questions about XR Technologies' Role in Society

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EXECUTIVE SUMMARY

In a thought experiment, the philosopher Robert Nozick (1974) proposed a hypothetical *experience machine* that allows people to live full and happy lives in virtual reality. If the user chooses so, they could plug themselves into the machine and live in a fully immersive, simulated world, better and more pleasurable than the real world. Once they enter this simulation, all memories of their past life are erased, and they will live the rest of their life in this space, believing it to be real. Nozick argued that most people would refuse the invitation to plug into the machine, and there is some empirical evidence to support his intuition (Hindriks & Douven, 2018; Weijers, 2014). While Nozick's thought experiment was specifically conceived to make the argument that humans desire more than just pure hedonism such as staying "in contact with reality" (Nozick, 1974, p. 646) and upholding social responsibility (Silverstein, 2000), the fictitious experience machine can serve as a useful illustration to identify the interests and concerns that users, onlookers, and stakeholders hold in regards to the immersive extended reality (XR) technologies.

In a sense, the name of the conference, Question Reality, reflects Nozick's thought experiment by stoking thoughts about XR's epistemological nature. In popular entertainment, sci-fi authors and script writers have explored the clashes between simulated virtual worlds and the real physical world (Humann, 2019). Fictions like *The Matrix* and *Ender's Game* explore themes tied to societies' fears that ever-advancing XR technologies can trap humans and create believable deceptions. By comparison, narratives such as *Ready Player One* juxtaposes a fun and colorful metaverse against the run-down neighborhood in the physical world to warn how VR technologies could harm people's sense of social responsibility for greater public needs.

In the present, contemporary virtual and augmented reality devices fall far short of the capabilities imagined by either Nozick or sci-fi writers. Rather, our current technologies are imperfect in that their affordances to produce fully believable illusionary worlds that could perfectly capture and trick the human mind are unrealized (and likely still be so in the foreseeable future). Today's XR devices can certainly invoke a feeling of presence in virtual spaces and with virtual entities (Lombard & Ditton, 1997), but technological and psychological constraints limit the creation of a true experience machine. From a technological standpoint, inconsistencies between the natural mapping on one's real body and their virtual avatar due to lag, glitches, etc., break our sense of spatial presence (Wirth et al., 2007). In virtual social interactions, non-human agents often do not fully appear or behave in ways that feel completely genuine (Nowak & Fox, 2018). For example, virtual agents that embody contemporary conversational AI have been described as unnatural "Socratic parrots" (Musi et al., 2025), which, while human-like, still have recognizable artificial patterns.

From a psychological standpoint, even if XR devices overcome their technological limitations to produce the perfect sensory illusion, they still would not be able to make users forget about the real world and, as such, people will use reality as a reference point and a comparison when navigating virtual worlds. This is because people's higher-order cognitive systems will override any quick, initial impressions of the virtual world's believability (Reeves & Nass, 1996), especially for those who are most experienced with the medium (Gambino et al., 2020). In other words, while



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virtual simulations can ‘feel’ real, most people understand that they are not. Nevertheless, even if people know and can verbalize that virtual spaces and entities are not physical and real, XR’s ability to immerse and elicit presence can still impact people’s thoughts, attitudes, and behaviors in important ways (e.g., Ahn, 2021).

With these considerations in mind, even if today’s technologies are not like sci-fi fictions or philosophers’ thought experiments, the comparison between the two can still be disruptive toward a general population’s understanding of XR’s uses and effects (Katins et al., 2025). One comparison of note is XR’s affordance to create *boundaries*. In its design, XR produces outputs that create a *private and individualized space* that is separate from others’ sensory environments. In simple terms, others do not get to see what the user sees when they are wearing a head-mounted device. Of course, the type of spaces and boundaries can vary—VR spaces create a clear distinction between virtual and real-worlds, whereas AR blurs the two realities together by overlaying virtual entities onto physical space. Nevertheless, many VR and AR spaces cannot be directly co-consumed with others close by (though there are exceptions like phone-based AR; Paasovaara et al., 2017). The inability for XR to be co-consumed with others in most situations originates from predecessors like stereoscope devices (e.g., View-Master) and kinetoscopes, where sensory outputs can only be directed to one person at a time. Similarly, this affordance overlaps to some degree with personal music players and earphones, where a person can consume music and podcasts in full privacy. However, it is much easier to allow co-consumption with music (by just sharing or unplugging one’s earphones) compared to XR.

If one has a rudimentary knowledge of XR, these boundaries may be perceived as a very isolating experience, and regarding the technology’s history and how it mirrors fictional depictions, this may certainly be true. For instance, while the augmented reality functions of the Meta Quest 3 allow users to see and interact with others in physical spaces, wearing the head-mounted device limits others’ ability to read nonverbal expressions from the user’s eyes and forehead. Likewise, unlike most traditional mediums, XR’s modality affordances (Sundar et al., 2015)—that immerse users into virtual environments through commandeering our senses of sight, hearing, touch, and position—by definition barricade people from others in the real world and vice versa. Moreover, if socializing is established inside the virtual spaces through online channels, the per-

ceived bandwidth for nonverbal expression is reduced due to the need to embody virtual avatars (Fox & McEwan, 2019; though advancements in facial mapping are starting to overcome this limitation; e.g., Vive, n.d.). Together, these examples display how the advantages of XR to create individualized immersive experiences can inadvertently restrict its perceived ability to facilitate shared and rich interpersonal experiences.

While socializing and bonding are rewarding aspects of interactive and non-interactive media (Grady et al., 2022), the zeitgeist of what virtual reality epitomizes feels far and distant from what a layman might consider a “social” medium (Katins et al., 2025). Its material and perceived barriers make it seem that XR’s mediated channel is not apt for communicating social interactions (compared to messaging apps or online video games) nor an appropriate target for sharing media experiences together (such as watching a movie during a date). Of course, experts of VR technology understand that the idea that modern head-mounted devices cannot be social is far from the truth (e.g., Schroeder, 2002). For instance, applications such as VRChat provide customizable virtual spaces where users can embody digital avatars and interact and play with others. Other XR applications are experimenting with virtual rooms designed for immersive co-viewing (e.g., Xtadium for sports spectatorship). Moreover, streaming XR gaming on sites like Twitch has become a popular social activity where thousands can watch a user play in virtual reality (Wu et al., 2023).

The discrepancy in the perceived strengths of XR’s social affordances between laymen and experts exemplifies key questions regarding its place in people’s day-to-day life and why it still feels very enigmatic and inaccessible to many. Most people have an idea of what virtual and augmented reality are, and science fiction speculates that it could have a major philosophical, ethical, psychological, and sociological impact, but it is also something that is rarely the focus of public discourse, especially compared to other emerging technologies such as social media and artificial intelligence. This mismatch raises many questions about XR’s relevance among scholars and professionals. Is it a technology that actually has a meaningful presence in society? Is the hype of a metaverse by Mark Zuckerberg and other technocrats (Meta, 2021) just a passing fad, or is it something that could truly revolutionize how we work and play, akin to the impact of smartphones (Katz, 2007)? Do XR’s relative disadvantages to other emerging media technologies limit its full adoption by the public (Davis, 1989; Rogers, 1962)? Will XR and/or future iterations of the technology create experiences that people actually want, and if so, can we accurately predict what those experiences will be?

BACKGROUND ON THE QUESTIONING REALITY CONFERENCE

These were the topics of discussion at the first Questioning Reality conference in 2024, hosted by the University of Toronto’s Data Science Institute. During this conference, experts in XR across disciplines and professions, including social scientists, computer scientists, and designers, participated in several blue-sky and brainstorming events to discuss the current and future role of XR in society. Notably, since advancements in XR continuously introduce new capabilities that may

not be evenly understood or experienced across scholars and professionals, several pre-conference activities were carried out where attendees could join a social VR space (e.g., VRchat) hosted by a conference fellow. During these pre-conference events, attendees interact with one another online and discuss broader implications of social VR within the space itself. At the in-person conference, attendees were invited to bring various head-mounted devices and VR applications (both commercial and developed in-house) from their labs to demo for others who may never have had the opportunity before. This approach to conferencing follows the traditions of Hiltz (1978), who believed that giving attendees the opportunity to use the medium of discussion would allow for cross-pollination of ideas between scholars and professionals who specialize across many different disciplines.

For instance, attendees were able to demo the Apple Vision Pro and its ability to use biometric data to create virtual avatars and to output 3D augmented movies. Due to its high cost at the time and its restrictions for customization, many attendees did not have firsthand experience with this specific product. Likewise, researchers from the Center for Advanced Human-Computer Ecosystems brought an in-house developed virtual world that simulated a storm surge off the U.S. coast and allowed users to practice mitigation techniques (Frank et al., 2025). This allowed attendees to learn about the ways VR could be utilized as a social intervention.

This white paper synthesizes the discussions during the blue skies and breakout sessions that occur between XR demoing, presentations, and panels. In particular, the paper highlights the major qualitatively coded themes that arose from these discussions. Table 1 lists some of the major themes and codes developed from the notes taken during the conference.

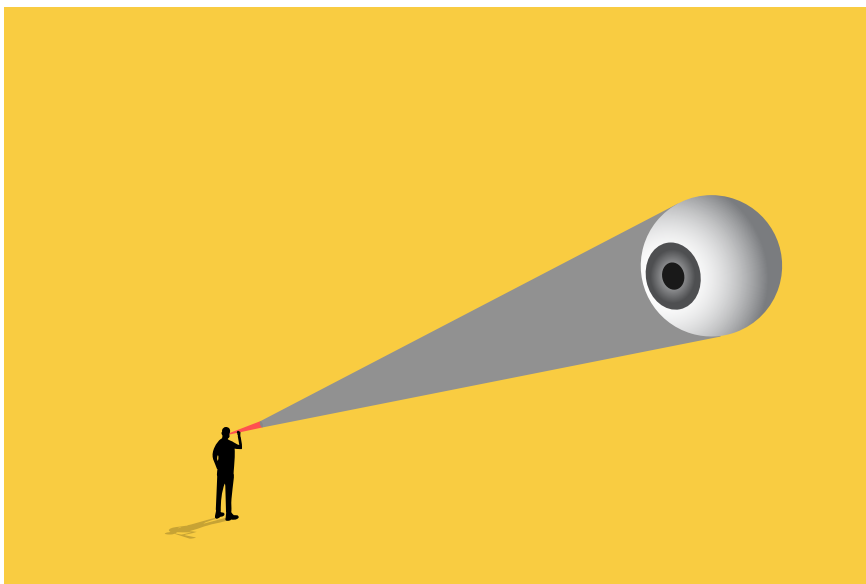
PHILOSOPHICAL APPROACHES/CONCERNS

The first set of major themes (*Philosophical Approaches & Concerns*) conveys XR's epistemological, ontological, and axiological nature. Mirroring the introduction of this white paper, questions were raised regarding whether XRs' virtual simulations are fantastical in nature (*Virtual Reality as Real*) and to what extent they are separated from the real-world and real-world consequences (*Physical Social Isolation; Ethics*). Both are interrelated; the affordances of nonphysicality, distinguishability of spaces, and ease of altering XR-mediated people and things may prompt a sense of perceived unrealisticness, which can shape real psychological and social outcomes, such as an increase in hyperpersonal- and disinhibition-related (Suler, 2004; Walther et al., 2015) behaviors. For instance, if immersive XR is perceived as a separate and non-consequential space rather than real mediated communication, will users use the space as a moral playground to try out antisocial behaviors that would not be appropriate in the real world (i.e., Suler, 2004)? If so, the immersive affordances of XR could encourage some users to engage in antisocial behaviors such as dehumanization, harassment, and sexual assault of others online (Raja & Al-Baghili, 2025). Moreover, since XR is an emerging technology, the absence of established norms (Lapinski & Rimal, 2005) and its usage by vulnerable populations (e.g., children) may heighten these concerns. This is in line with recent scholarly thoughts (Slater et al., 2020). Moreover, many of the attendees in the conference brought up other very real consequences that XR technologies may

Table 1

Sample of the Major Themes and Codes Developed from Notes Taken during Questioning Reality '24

Sample of Major Themes	Sample of Codes
Philosophical Approaches/Concerns	<i>Virtual Reality as Real; Ethics; Physical Social Isolation; Data/Social Privacy; Harassment and SA in VR; Pervasive Tech</i>
Barriers to Adoption/Routes to Adoption	<i>Slow Diffusion; Price; Unsure of Function; Difficult to Use; Low Reward; Niche; Improve Components; Use Case; Amazing Things</i>
Social VR Affordances	<i>Presence; Personalization; Interactivity; Portability; Bandwidth; Navigability</i>
Technical Improvements/Future Possibilities	<i>AI (LLM) Integration; Digital Economy; Collaboration; Overcome the Physical Self; Disentrainment - Physical Space</i>



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promote, including concerns about digital privacy and surveillance; echo chambers and polarization, and the digital divide between people with and without access to the technology.

On the other hand, we should also examine the possible ethical and social consequences in light of broader cycles of public sentiment (Orben, 2020) in which emerging technologies are often seen with suspect eyes. As such, the conference also speculated on the idea that if XR technologies become

pervasive (*Pervasive Tech*), then what good things can it do (*Ethics*), and what things can the technology not solve (*Asking too much of Tech*)? For instance, regarding the perceived realism of the virtual world, the Proteus effect suggests that people will begin to mimic characteristics of their believable embodied avatar (Yee & Bailenson, 2007). Could users leverage this effect to maintain

and promote their well-being? Similarly, greater empathy can emerge from embodying avatars different from oneself (Ahn et al., 2013), and this may serve as an important prosocial consequence that could promote social well-being.

To sum, while the philosophical debate on whether virtual reality is real or not real (and the larger question of how the label 'real' is ontologically defined) may be up in the air, there is little doubt that, as a mediated channel, XR creates real outcomes that scholars and practitioners should consider. All digitally mediated communication has physical, psychological, social, and monetary consequences that have important implications for how the technology should be developed and used.

BARRIERS TO ADOPTION/ROUTES TO ADOPTION

If one is familiar with XR's history, it would not be a surprise that the technology has not traditionally been successful in the marketplace. As such, much of the discussion during the conference centered on the why and why not people will adopt and use XR technologies (*Barriers to Adoption & Routes to Adoption*). There are certainly well-known products in the past, like Nintendo's Virtual Boy and Google Glass, that were commercial failures. Even today, with VR adoption growing, it is nothing compared to the growth of other emerging media technologies (e.g., smartphones; *Slow Diffusion*). One of the predominant conversations during the conference asked why this is the case. Certainly, there has been on-and-off hype for the technology for decades (e.g., Meta, 2021), but the public response has been lukewarm.

XR's lackluster adoption and usage can also be attributed to its inherent design. Advancements in media selection research have suggested that usage will be determined by the relative weighing of its benefit versus its cost (Gong & Husky, 2024). From a diffusion standpoint (Rogers, 1962), XR technologies are difficult to observe and assess their advantages and disadvantages (*Unsure of Function*). Moreover, from a demand perspective (Bowman, 2018), XR technologies have a high learning curve for setup, using the controls and interface, and navigating virtual worlds. XR is also often not very comfortable, and most users can only handle a limited amount of time inside a head-mounted device. Together, this often requires a level of physical and cognitive effort that is not worthwhile for many people (*Difficult to Use*). As such, the benefits and flexibility of competing media outweigh any pros in using XR day-to-day (*Low Reward & Displaces Other Tasks*). For example, a person cannot multitask with their phone or other media when in VR. Similarly, given that the primary use of XR technologies is for entertainment, the XR hesitation may also be driven by guilt that one's time could be better utilized for other non-entertainment activities (Reinecke et al., 2014). Lastly, there are higher monetary and social costs that create a digital divide between adoptees and non-adoptees (*Price*).

Certainly, there is room for XR to improve that may bolster usage. Attendees at the conference indicated that decreasing the weight/size, creating easier interfaces (*Improve Components*), having an increased focus on content, and tailoring to specific populations may help the technology in the future. For example, some research indicates that VR uses differ between individuals who

are more or less sensitive to experiencing presence (Cummings et al., 2022). Therefore, focusing on technological advancements and marketing tailored for niche (rather than wide) audiences may be more appropriate (*Niche*). Others at the conference highlighted XR's unique content and experiences as something that should foster greater usage if uncertainty about the technology is reduced. These include content where first-hand experiences are inaccessible or impossible (e.g., virtual traveling) and experiences that are hyper-realistic (*Amazing Things*). In a similar vein, beyond entertainment and leisure, identifying XR's strengths for learning (e.g., through digital twins) and persuasion (Ahn, 2021) may help bring the technology into public and private spaces where it may be used for training and risk mitigation (Frank et al., 2025) (*Use Case*). Finally, some emphasized the importance of social connections for XR adoption and use. In order to have a successful social platform on these devices, a critical mass of users needs to be established first.

SOCIAL VR AFFORDANCES

Social factors (social needs, affordances, and environments) are central to people's media usage and reception (Grady et al., 2022). Building upon the previous themes, the conference hosted a discussion on what makes XR technologies social, specifically their social affordances. While the general public typically does not view prototypical XR devices as social (Katins et al., 2025), scholars and practitioners are beginning to emphasize how technological affordances encourage people to interact with each other and with other embodied entities (Schroeder, 2002). Certainly, XR applications such as Pokémon Go are highly social by allowing people to co-play the AR game together (Paasovaara et al., 2017), and users of head-mounted devices develop creative methods to circumvent the barriers of having a device covering one's face (Wu et al., 2023). In this sense, a greater understanding of what affordances emerge from the interactions between users and design features that allow and restrict social XR (and especially social VR) is warranted. Computer-Mediated Communication and Human-Machine Communication researchers have pinpointed important affordances that shape interpersonal and mass-personal interactions within emerging technologies (Fox & McEwan, 2019; Sundar et al., 2015). Some of these include personalization, interactivity, portability, bandwidth, and navigability. For instance, in a social VR environment, the ability to personalize one's avatar may help facilitate self-expression and creativity, whereas using cues to prompt users to understand spatial location and movement may help facilitate which social spaces



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are appropriate or not for certain activities and interactions. Moreover, engaging in multimedia interactions (i.e., using multiple channels for communication) may also be useful to help overcome some of XR's design limitations (e.g., isolating barriers). For example, it is sometimes easier to offload some of the communication needs to other channels (e.g., FaceTime) when setting up and coordinating social XR platforms between people in different physical locations (Baldwin et al., 2023). Together, attendees recognized the importance of social affordances for this emerging media and the need to examine how these affordances can be better communicated.

TECHNICAL IMPROVEMENTS/ FUTURE POSSIBILITIES

The last theme this white paper will note regards XR's future (*Technical Improvements & Future Possibilities*). During the conference, attendees were given the opportunity to speculate about the future of VR and what the technology, its contents, and its role in society might look like. While thoughts on the topic ranged from the highly feasible to edging on unlikely speculations, these conversations nevertheless underscore the technology's design features and consequences that hover in experts' minds.

From the hardware side, some predicted that VR technologies would have improved components (e.g., Moore's law; Moore, 1965) that would allow for more photorealistic outputs and greater immersion. The technology will also become lighter and more wearable to allow for easier day-to-day usage (Nass & Mason, 1990), and it will converge and connect with other mediums more seamlessly. For example, it may be easier for a user to have their avatar be portable by connecting VR devices with smartphones. From the content side, attendees predicted that format conventions will become solidified, especially the presentation of narrative content. By comparison, they also predicted an increase in participatory content creation. With the integration of generative AI and open source repositories, users will have more opportunities to build tailored virtual environments using self-generated digital assets and agents in order to achieve their unique desires and goals (*AI (LLM) Integration*). This aligns with developing research into mass self-communication, wherein scholars have begun to examine the effect of media created by oneself (Valkenburg et al., 2016).

With these advancements, attendees also speculated about the state of the real-world in light of an ever increasing digital world. Would there be a disenchantment of physical spaces? For instance, could classes be taught completely in virtual environments with the help of AI-embodied agents (*Disenchantment - Physical Space*)? If so, would this remove the need for physical schools? Moreover, if immersive virtual spaces become a daily place to work, hang out, and socialize, will the norms that develop differ from the norms in "reality?" Certainly, there are some norm differences (and the language we use to communicate these norms) between the real world and interacting on social media today (e.g., doxxing). Perhaps, a similar shift in norms and language may also evolve. Together, while predicting which particular technological developments will actually manifest and popularize is an elusive task, we can rely on the robust theoretical and practical groundworks to understand the human in the machine, if and when these developments arise.

CONCLUSION

In comparison to Nozick's (1974) experience machine (and the myriad of sci-fi that portray simulated virtual experiences), today's XR machines are "imperfect" in creating simulations that everyone wants to live in. The inaugural Questioning Reality conference highlighted this tension between the types of realities people might want to experience, under what conditions, and for how long. In part of these conversations, a central theme focused on the spatial and social aspects of XR, whether it is interacting with virtual agents, other users through avatars, or people outside in one's immediate physical space. The general public understanding is that VR is a closed-door room—isolating and personal (Katins et al., 2025), even if this is not a hard truth (Schroeder, 2002). Therefore, our continuing discussion on the role XR does or does not play in people's lives should give special attention to how virtual environments create spaces that include and exclude others and affordances that can arise from technological advancements as well as the potential benefits and harms that may be consequences of these affordances.

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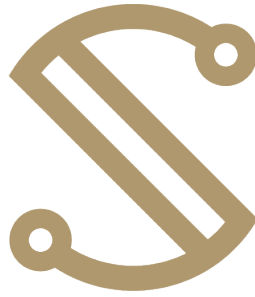
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